

Installation Instructions

SmartSociety — Web-Based Housing Society Management System

Target Local development machine

Date August 2026

1 Prerequisites

REQUIREMENT	NOTES
Node.js 18+	Backend and frontend both run on Node
npm	Ships with Node
MySQL / MariaDB	XAMPP's bundled MariaDB works as-is — no separate MySQL install needed

2 Database Setup

1 Start MySQL

If using XAMPP, open the XAMPP Control Panel and start the **MySQL** module.

2 Create the database

Using phpMyAdmin or the `mysql` CLI:

```
CREATE DATABASE smart_society;
```

3 Import the schema (optional)

The table structure — including the audit-log immutability triggers — is checked in at `Backend/database/smart_society_schema.sql`. Importing it is optional: the backend also creates any missing tables automatically on first boot via `sequelize.sync()`, but importing the file up front also installs the same triggers immediately and matches exactly what was tested.

```
mysql -u root smart_society < Backend/database/smart_society_schema.sql
```

3 Backend Setup

1 Install dependencies

```
cd Backend  
npm install
```

2 Configure environment variables

Copy `.env.example` to `.env` and fill in the database credentials (XAMPP's defaults — host `127.0.0.1`, user `root`, empty password — work without changes). See §5 for what each variable does.

```
cd Backend  
cp .env.example .env
```

3 Seed the first accounts

The API has no public route to create the first Admin or Guard, so the bootstrap script inserts them directly. Safe to re-run — it skips anything that already exists.

```
npm run seed
```

4 Start the server

```
npm run dev
```

Runs on `http://localhost:5000` by default. Look for `MySQL Connected` and `Server running on http://localhost:5000` in the console.

4 Frontend Setup

1 Install dependencies

```
cd frontend  
npm install
```

2 Configure the API URL

`frontend/.env` already points at the default backend port:

```
VITE_API_BASE_URL=http://localhost:5000/api
```

3 Start the dev server

```
npm run dev
```

Runs on `http://localhost:5173`. Open it in a browser and log in with any of the seeded accounts (see the User Credentials document).

5 Environment Variables Reference

VARIABLE	REQUIRED	PURPOSE
<code>PORT</code>	No — default 5000	Backend HTTP + WebSocket port
<code>DB_HOST</code> , <code>DB_PORT</code> , <code>DB_NAME</code> , <code>DB_USER</code> , <code>DB_PASSWORD</code>	No — default to XAMPP's local MariaDB	MySQL connection
<code>JWT_SECRET</code>	Yes	Signs login tokens — use a long random string
<code>CLOUDINARY_*</code>	No	Complaint photo uploads; falls back to local disk storage under <code>Backend/uploads/</code> if unset
<code>EMAIL_HOST</code> , <code>EMAIL_PORT</code> , <code>EMAIL_USER</code> , <code>EMAIL_PASS</code>	No	OTP + password-reset email; if unset, the server auto-provisions a throwaway Ethereum test inbox and returns a preview link in the API response so the flow still works out of the box
<code>FRONTEND_URL</code>	No — default <code>http://localhost:5173</code>	Used to build the password-reset link sent by email
<code>VITE_API_BASE_URL</code>	Yes (frontend)	Base URL the SPA sends API requests to

WITHOUT REAL SMTP CREDENTIALS

Login OTPs and password-reset links still work — they're delivered to an auto-created Ethereum inbox, and the API response includes a `devPreviewUrl` the frontend surfaces directly so the flow can be completed without checking a real inbox. This is a development convenience only; a production deployment should set real `EMAIL_*` values.

6 Verifying the Install

1 Backend health check

Visit `http://localhost:5000/health` — should return a JSON success message.

2 Log in

Open `http://localhost:5173`, log in as `Admin` / `admin123`, and complete the OTP step using the console-logged preview link (or real inbox if SMTP is configured).

3 Confirm real-time works

From the footer, hold the emergency-siren control to confirm — the banner should appear immediately on any other open, logged-in tab, proving the Socket.io channel is live.